Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

WHAT IT CALCULATES	RELATION
Design flow from the heat carried, when a heat load is given in place of a flow	$Q = \Phi/(\rho\, c_p\, \Delta T)$
Mean velocity in the pipe	$v = Q/(\pi D^2/4)$
Reynolds number, which decides laminar or turbulent flow	$Re = \rho v D/\mu$
Friction factor for laminar flow (Re < 2300)	$f = 64/Re$
Friction factor for turbulent flow, Colebrook–White by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it	$1/\sqrt{f} = -2\log_{10}(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$
Head lost in the pipe and its fittings	$h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$
System curve, with friction scaled from the design flow as Q^2 (a closed circuit has $\Sigma dz = 0$, the rise in the flow being recovered in the return)	$H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_f(Q)$
Water held in the pipes	$V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$
Duty point, where the pump and system curves cross (found by bisection)	$H_{pump}(Q) = H_{sys}(Q)$
Pump affinity laws for a change of speed n or impeller diameter D	$Q \propto n\, D^3$; $H \propto n^2\, D^2$
Net positive suction head available, to compare with what the pump requires	$NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8); Flow of fluids in pipes and ducts (principles)[3](Section 4); Density of liquid water[4].

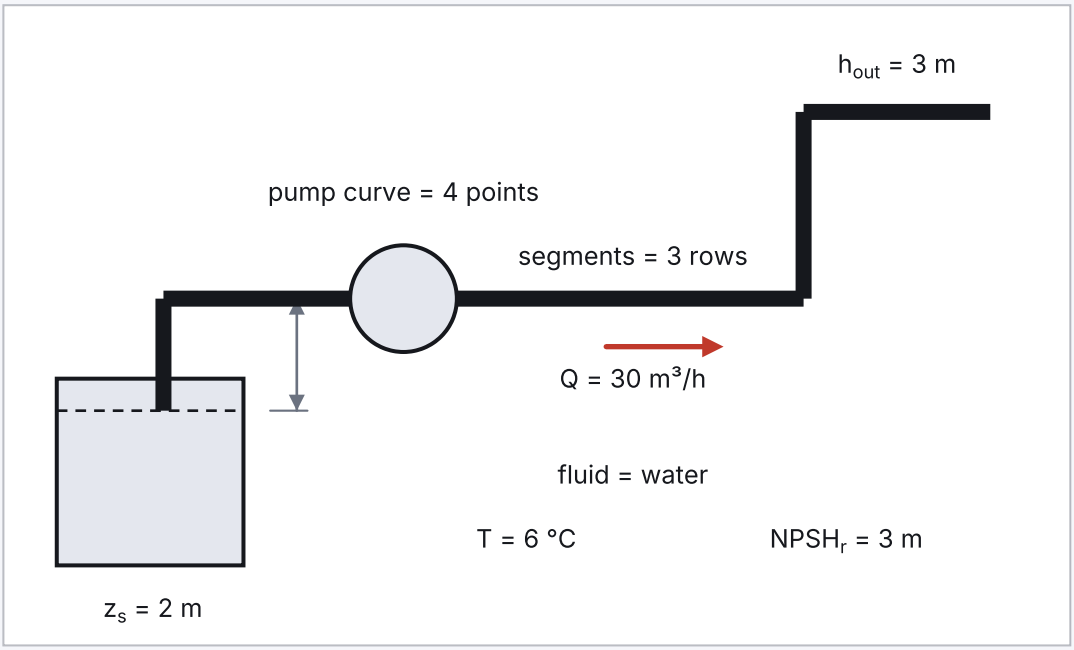


Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	`water`		spec: SPEC-M-04 §3: chilled water, no glycol	any
T	Fluid temperature (mean; link B6 t_m for a heating loop)	6	°C	spec: SPEC-M-04 §3: 6 °C flow	0 … 100 °C
Q	Design flow rate (blank when a heat load gives it)	30	m³/h	spec: Schedule CHW-01 design flow	0.01 … 100000 m³/h
segments	Pipe segments in series	`3 row(s)`		drawing: DRG-M-110 rev C	any
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	`open (the pump lifts the water by the rises entered)`		default	any
h_{out}	Pressure head required at the outlet	3	m	spec: Terminal unit: 30 kPa ≈ 3 m	0 … 1000 m
v_{max}	Velocity limit (warning)	2.5	m/s	spec: CIBSE Guide C practice limit, assumed	0.5 … 10 m/s
Q_{pump}	Pump curve: flow points	`[0, 20, 40, 60]`	m³/h	spec: Pump datasheet P-01 rev 2, curve points	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
H_{pump}	Pump curve: head points	[40, 36, 24, 4]	m	spec: Pump datasheet P-01 rev 2, curve points	any
n_{ratio}	Speed ratio n_2/n_1 (affinity)	0.9		assumption: Inverter turn-down to 90 % speed.	0.2 … 2
D_{ratio}	Impeller diameter ratio D_2/D_1 (affinity)	1		assumption: Impeller unchanged.	0.7 … 1.1
η_{pump}	Pump efficiency at duty (for shaft power)	0.68		spec: Pump datasheet P-01 rev 2	0.2 … 0.95
p_{atm}	Atmospheric pressure (NPSH)	101.325	kPa	assumption: Sea level.	50 … 110 kPa
z_s	Static head at the pump suction (+ flooded, − lift)	2	m	drawing: DRG-M-110 rev C	-10 … 100 m
$NPSH_r$	NPSH required by the pump (optional)	3	m	spec: Pump datasheet P-01 rev 2	0 … 50 m

TABLE ROW	VALUES
segments[1]	name = plantroom suction; L = 8 m; D = 100 mm; eps = 0.045 mm; K = 3; L_eq = 0 m; dz = 0 m; side = suction
segments[2]	name = riser DN80; L = 60 m; D = 80 mm; eps = 0.045 mm; K = 8; L_eq = 0 m; dz = 10 m; side = discharge
segments[3]	name = index run DN65; L = 45 m; D = 65 mm; eps = 0.045 mm; K = 12; L_eq = 6 m; dz = 2 m; side = discharge

3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with Q^2 from the design flow (turbulent assumption); the static part is independent of flow. ($H_{sys} = H_{static} + k\,Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. n_{ratio} (Speed ratio n_2/n_1 (affinity)) is taken as 0.9. (Inverter turn-down to 90 % speed.)
A6. D_{ratio} (Impeller diameter ratio D_2/D_1 (affinity)) is taken as 1. (Impeller unchanged.)
A7. p_{atm} (Atmospheric pressure (NPSH)) is taken as 101.325 kPa. (Sea level.)

4. Working

[3] ρ = Kell (1975) density of water at T [4]
= T = 6 °C
= 999.94 kg/m³
Density
[4] μ = Vogel fit μ = 0.02939 exp(507.88/(T_K − 149.3)) mPa·s
= T_K = 279.15 K
= 1.4684 mPa·s

Dynamic viscosity

[5] p_v = Hyland-Wexler saturation pressure
= T = 6 °C
= 0.93525 kPa

Vapour pressure (NPSH)

[6] T_{line} = T (no T_{line} entered)
= 6 °C
= 6 degC

Design temperature of the line to insulate: the fluid temperature T, since no T_{line} was entered

[10] Q_{design} = Q
= 30 m³/h
= 30 m³/h

Design flow (as entered)

[11] v_{s1} = plantroom suction: Q / (π D²/4)
= 30 m³/h / (π × 100² mm² / 4)
= 1.061 m/s

Velocity, plantroom suction

[12] Re_{s1} = plantroom suction: ρ v D / μ
= 999.94 × 1.061 × 0.1 / 0.0014684
= 72253

Reynolds number, plantroom suction

[13] f_{s1} = plantroom suction: Darcy friction factor
= Colebrook-White, 10 iterations (Swamee-Jain start 0.021152)
= 0.021093

Friction factor, plantroom suction

[14] $f_{sj,s1}$ = plantroom suction: Swamee-Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.021152

Explicit cross-check, plantroom suction

[15] $h_{f,s1}$ = plantroom suction: f (L + L_{eq})/D · v²/2g + K v²/2g
= 0.021093 × (8 + 0) / 0.1 × 0.057399 + 3 × 0.057399
= 0.26906 m

Head loss, plantroom suction

[16] v_{s2} = riser DN80: Q / (π D²/4)
= 30 m³/h / (π × 80² mm² / 4)
= 1.6579 m/s

Velocity, riser DN80

[17] Re_{s2} = riser DN80: ρ v D / μ
= 999.94 × 1.6579 × 0.08 / 0.0014684
= 90316

Reynolds number, riser DN80

[18] f_{s2} = riser DN80: Darcy friction factor
= Colebrook-White, 10 iterations (Swamee-Jain start 0.020946)
= 0.020849

Friction factor, riser DN80

[19] $f_{sj,s2}$ = riser DN80: Swamee-Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020946

Explicit cross-check, riser DN80

[20] $h_{f,s2}$ = riser DN80: f (L + L_{eq})/D · v²/2g + K v²/2g
= 0.020849 × (60 + 0) / 0.08 × 0.14014 + 8 × 0.14014
= 3.3123 m

Head loss, riser DN80

[21] v_{s3} = index run DN65: Q / (π D²/4)
= 30 m³/h / (π × 65² mm² / 4)
= 2.5113 m/s
Velocity, index run DN65
[22] Re_{s3} = index run DN65: ρ v D / μ
= 999.94 × 2.5113 × 0.065 / 0.0014684
= 111160
Reynolds number, index run DN65
[23] f_{s3} = index run DN65: Darcy friction factor
= Colebrook-White, 9 iterations (Swamee-Jain start 0.020964)
= 0.020834
Friction factor, index run DN65
[24] $f_{sj,s3}$ = index run DN65: Swamee-Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020964
Explicit cross-check, index run DN65
[25] $h_{f,s3}$ = index run DN65: f (L + L_{eq})/D · v²/2g + K v²/2g
= 0.020834 × (45 + 6) / 0.065 × 0.32155 + 12 × 0.32155
= 9.1151 m
Head loss, index run DN65
[26] $h_{f,total}$ = Σ h_f
= 3 segment(s)
= 12.696 m
Total friction and fitting loss at the design flow
[28] V_{pipe} = Σ π/4·D²·L
= 3 segment(s)
= 513.75 L
Water held in the pipe segments
[29] H_{static} = Σ dz + h_{out}
= 12 + 3
= 15 m
Static head
[30] H_{sys} = H_{static} + $h_{f,total}$
= 15 m + 12.696 m
= 27.696 m
System head at the design flow
[31] dp_{total} = ρ * 9.80665 m/s^2 * $h_{f,total}$
= 999.94 kg/m³ × 9.80665 m/s^2 × 12.696 m
= 124.5 kPa
Friction pressure drop at the design flow
[32] dp_{sys} = ρ * 9.80665 m/s^2 * H_{sys}
= 999.94 kg/m³ × 9.80665 m/s^2 × 27.696 m
= 271.59 kPa
System pressure at the design flow
[33] P_{hyd} = ρ * 9.80665 m/s^2 * Q_{design} * H_{sys}
= 999.94 kg/m³ × 9.80665 m/s^2 × 30 m³/h × 27.696 m
= 2.2633 kW
Hydraulic power at the design point
[39] Q_{duty} = duty flow (as entered)
= $H_{pump}(Q)$ = $H_{sys}(Q)$, bisection
= 31.569 m³/h
Duty flow as entered

[40] H_{duty} = duty head (as entered)
= $H_{sys}(31.569\ m^3/h)$
= 29.059 m
Duty head as entered
[41] $P_{shaft} = \rho\ g\ Q\ H\ /\ \eta$
= 999.94 × 9.8066 × 0.008769 × 29.059 / 0.68
= 3.6747 kW
Shaft power at duty as entered
[42] affinity = $Q_2 = Q_1\ (n_2/n_1)(D_2/D_1)^3$, $H_2 = H_1\ (n_2/n_1)^2(D_2/D_1)^2$
= $n_2/n_1 = 0.9$, $D_2/D_1 = 1$
= "curve scaled"
Affinity laws applied to the pump curve
[43] Q_{duty2} = duty flow (after the affinity shift)
= $H_{pump}(Q) = H_{sys}(Q)$, bisection
= 26.238 m³/h
Duty flow after the affinity shift
[44] H_{duty2} = duty head (after the affinity shift)
= $H_{sys}(26.238\ m^3/h)$
= 24.712 m
Duty head after the affinity shift
[45] $P_{shaft2} = \rho\ g\ Q\ H\ /\ \eta$
= 999.94 × 9.8066 × 0.0072883 × 24.712 / 0.68
= 2.5972 kW
Shaft power at duty after the affinity shift
[48] $h_{f,suction}$ = Σ h_f on the suction side
= 1 suction segment(s)
= 0.26906 m
Suction-side friction loss
[49] $NPSH_a = p_{atm}$ / (ρ * 9.80665 m/s^2) + z_s - $h_{f,suction}$ - p_v / (ρ * 9.80665 m/s^2) (Ch. 8)
= 101.32 kPa / (999.94 kg/m³ × 9.80665 m/s^2) + 2 m − 0.26906 m − 0.93525 kPa / (999.94 kg/m³ × 9.80665 m/s^2)
= 11.968 m
NPSH available
[51] $u_{npsh} = NPSH_r\ /\ NPSH_a$
= 3 m / 11.968 m
= 0.25066
NPSH utilisation (required / available)
[52] v_{peak} = highest segment velocity
= 2.5113 m/s
= 2.5113 m/s
Highest velocity
[53] $u_v = v_{peak}\ /\ v_{max}$
= 2.5113 / 2.5
= 1.0045
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
T_{line}	6 degC	Line design temperature
v_{s1}	1.061 m/s	Velocity, plantroom suction
Re_{s1}	72253	Re, plantroom suction
f_{s1}	0.021093	f, plantroom suction

SYMBOL	VALUE	DESCRIPTION
$f_{sj,s1}$	0.021152	f (Swamee–Jain), plantroom suction
$h_{f,s1}$	0.26906 m	h_f, plantroom suction
v_{s2}	1.6579 m/s	Velocity, riser DN80
Re_{s2}	90316	Re, riser DN80
f_{s2}	0.020849	f, riser DN80
$f_{sj,s2}$	0.020946	f (Swamee–Jain), riser DN80
$h_{f,s2}$	3.3123 m	h_f, riser DN80
v_{s3}	2.5113 m/s	Velocity, index run DN65
Re_{s3}	111160	Re, index run DN65
f_{s3}	0.020834	f, index run DN65
$f_{sj,s3}$	0.020964	f (Swamee–Jain), index run DN65
$h_{f,s3}$	9.1151 m	h_f, index run DN65
$h_{f,total}$	12.696 m	Σh_f
H_{static}	15 m	Static head
V_{pipe}	513.75 L	Water held in the pipes
H_{sys}	27.696 m	System head at Q
dp_{total}	124.5 kPa	Friction Δp
dp_{sys}	271.59 kPa	System Δp
Q_{design}	30 m³/h	Design flow
P_{hyd}	2.2633 kW	Hydraulic power
Q_{duty}	31.569 m³/h	Duty flow as entered
H_{duty}	29.059 m	Duty head as entered
P_{shaft}	3.6747 kW	Shaft power as entered
Q_{duty2}	26.238 m³/h	Duty flow after the affinity shift
H_{duty2}	24.712 m	Duty head after the affinity shift
P_{shaft2}	2.5972 kW	Shaft power after the affinity shift
$NPSH_a$	11.968 m	NPSH available
v_{peak}	2.5113 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Velocity ≤ limit	u_v = 1.0045	u_v ≤ 1	1	
PASS	$NPSH_r \le NPSH_a$	u_{npsh} = 0.25066	u_{npsh} ≤ 1	0.251	
PASS	Duty point exists (as entered)	31.569 m³/h	29.059 m	0	

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Duty point exists (after the affinity shift)	26.238 m³/h	24.712 m	0	

7. Applicability

WARNING — input.circuit: circuit (Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)) was not entered; the template default 'open (the pump lifts the water by the rises entered)' is used.
WARNING — velocity: segments: index run DN65 runs at 2.5113 m/s, above the 2.5 m/s limit.

8. Verification

```
Template bs.pipe-pump@1.6.0 Content hash 5a30160d8816872d Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.